Geometric Unification of Flavor: Masses, Mixing, and CP from the Golden Ratio Lattice

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Abstract

We present a unified geometric framework for Standard Model flavor: fermion masses, mixing matrices, and CP-violating phases are consistent with arising from the structure of a single arithmetic lattice embedded in a symmetric space. The key observation is that the observed fermion masses cluster near the discrete set

$$m = m_e \cdot \phi^{q/4}, \qquad q \in \mathbb{Z},$$

where $\phi = (1+\sqrt{5})/2$ is the golden ratio and m_e the electron mass. This lattice is precisely the image of the logarithmic unit lattice of $\mathbb{Q}(\sqrt{5})$ under the exponential map of the symmetric space $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ (the hyperbolic plane), with covolume $\log \phi$. The quarter-integer spacing arises from the metric normalization of the maximal flat.

We demonstrate that the same geometric construction extends to mixing and CP violation via the Iwasawa decomposition $\mathrm{PSL}(2,\mathbb{C}) = KAN$ (Figure 4): the K- and N-factors encode the CKM and PMNS mixing matrices (companion papers), while the A-factor (twist angles) gives CP phases that are holonomies of flat U(1) connections classified by the torsion $H_1(M) = \mathbb{Z}/5$ of the underlying hyperbolic 3-manifolds. The entire flavor sector is thus encoded in the discrete geometry of a single arithmetic lattice and its associated symmetric space.

All results are presented under a strict methodological contract: fixed conventions, bounded scans, explicit falsifiability criteria. Residuals are stable and small; null tests confirm non-random clustering. The framework provides a geometric origin for the golden ratio in particle physics and unifies masses, mixing, and CP in a single arithmetic-geometric structure.

1 Introduction

The flavor parameters of the Standard Model"'fermion masses, mixing angles, and CP phases"'are among the most precisely measured quantities in particle physics, yet their origin remains unexplained. The observed mass hierarchy spans more than twelve orders of magnitude, and the mixing matrices exhibit a characteristic pattern often described as "Cabibbo suppression" and "weak mixing." Many theoretical approaches invoke discrete flavor symmetries, extra dimensions, or string-inspired constructions, but they typically introduce free parameters and lack a unifying geometric principle.

In this work we propose that the entire flavor structure is consistent with being encoded in a single arithmetic lattice embedded in a symmetric space. The lattice is the logarithmic unit lattice of the quadratic field $\mathbb{Q}(\sqrt{5})$, which appears naturally in the geometry of the symmetric space $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ (the hyperbolic plane). Its covolume is the regulator $\log\phi$, where $\phi = (1+\sqrt{5})/2$ is the golden ratio. Under the exponential map of the symmetric space, this lattice gives the discrete set

$$m = m_e \cdot \phi^{q/4}, \qquad q \in \mathbb{Z},$$

where the quarter-integer spacing emerges from the metric normalization of the maximal flat.

Remarkably, this set coincides with the observed fermion masses: the electron defines $q = 0$, the muon and tau cluster near $q = 44$ and $q = 68$, the light quarks near $q = 12, 18, 43$, the heavy quarks near $q = 65, 75, 106$, and the W, Z, and Higgs bosons near $q = 99, 101, 103$ (Table 1). The residuals are small, typically a few percent in mass ratio, and the pattern is stable under perturbations of input values.

Beyond masses, the same geometric framework accommodates mixing and CP violation. The Iwasawa decomposition $\mathrm{PSL}(2,\mathbb{C}) = KAN$ of the holonomy group of a compact hyperbolic 3"manifold provides a natural split: the K"factor (maximal compact) gives the CKM quark mixing matrix, the N"factor (unipotent) gives the PMNS lepton mixing matrix, and the A-factor (diagonal) gives CP phases as flat U(1) holonomies. The underlying manifolds"'$m006$ for quarks and $m003$ for leptons"'both have first homology $H_1 = \mathbb{Z}/5$, which quantizes the CP phase to a fifth root of unity, matching the observed Jarlskog invariant. The connection between the mass lattice and the mixing sector (Figure 1) is indirect but unified: both stem from the same arithmetic hyperbolic geometry.

We identify the precise arithmetic mechanism linking the two sectors. The torsion $H_1(M) = \mathbb{Z}/5$ classifies flat U(1) connections on M via characters $\chi : \mathbb{Z}/5 \to \mathrm{U}(1)$, giving holonomies $e^{2\pi ik/5}$ — fifth roots of unity. These generate the cyclotomic field $\mathbb{Q}(\zeta_5)$, whose maximal real subfield is $\mathbb{Q}(\zeta_5+\zeta_5^{-1}) = \mathbb{Q}(\sqrt{5})$, with fundamental unit $\phi = (1+\sqrt{5})/2$. The regulator of $\mathbb{Q}(\sqrt{5})$ is $\log\phi$, which is precisely the lattice spacing of the mass formula. In particular, $2\cos(2\pi/5) = \phi - 1 = 1/\phi$, so the CP phase $e^{2\pi i/5}$ and the golden ratio ϕ are two facets of the same $\mathbb{Z}/5$ structure. The entire flavor sector is thus tied to the single arithmetic datum $H_1(M) = \mathbb{Z}/5$.

This paper is structured as a methodological and empirical synthesis. Section 2 recalls the symmetric-space construction and derives the mass lattice. Section 3 describes the encoding, anchoring, scan protocol, and null tests in detail. Section 4 presents the mass clustering results. Section 5 outlines the extension to mixing and CP violation, referencing companion papers. Section 6 discusses the spectral properties of the underlying manifolds. Section 10 discusses falsifiability, interpretive limits, and future directions. Section 12 concludes.

2 Geometric foundation: the golden ratio lattice

The key geometric input comes from the theory of symmetric spaces and arithmetic groups. The space of positive definite quadratic forms modulo overall scale is the symmetric space

$$\mathcal{M}_n = \mathrm{SL}(n, \mathbb{R})/\mathrm{SO}(n)$$

(see [2] for a detailed exposition). Its tangent space at the identity is the space of traceless symmetric matrices. A maximal flat is obtained by exponentiating the Cartan subalgebra of diagonal matrices with zero trace, which is isometric to $\mathbb{R}^{n-1}$ with the Euclidean metric $\langle X, Y\rangle = \mathrm{tr}(XY)$.

For $n = 2$, $\mathcal{M}_2 \cong \mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2)$ is the hyperbolic plane. Its maximal flats are geodesics. The real quadratic field $K = \mathbb{Q}(\sqrt{5})$ has unit group $\mathcal{O}_K^\times \cong \{\pm 1\} \times \mathbb{Z}$. The logarithmic embedding maps units to vectors in $\mathbb{R}^2$:

$$\varepsilon \longmapsto (\log|\sigma_1(\varepsilon)|,\ \log|\sigma_2(\varepsilon)|),$$

where $\sigma_{1,2}$ are the two real embeddings. Because $N_{K/\mathbb{Q}}(\varepsilon) = \sigma_1(\varepsilon)\sigma_2(\varepsilon) = \pm 1$, the image lies in the line $x_1 + x_2 = 0$, which can be identified with the Lie algebra $\mathfrak{a} = \mathrm{Diag}_0(2, \mathbb{R}) = \{\mathrm{diag}(t, -t) : t \in \mathbb{R}\}$.

The fundamental unit $\varepsilon_0 = \phi = \frac{1+\sqrt{5}}{2}$ gives $\Lambda(\phi) = (\log\phi, -\log\phi)$. Its covolume in $\mathfrak{a}$ (the regulator) is $R_K = \log\phi$.

Under the exponential map $\exp : \mathfrak{a} \to \mathcal{M}_2$, the lattice $\mathbb{Z} \cdot \Lambda(\phi)$ maps to the discrete set

$$\left\{\exp\bigl(t\,\mathrm{diag}(1, -1)\bigr) : t = k\log\phi\right\} = \left\{\mathrm{diag}(\phi^k, \phi^{-k})\right\}.$$

The distance from the identity $\mathrm{diag}(1, 1)$ to $\mathrm{diag}(\phi^k, \phi^{-k})$ is $\sqrt{2}\,k\log\phi$. To obtain a lattice with spacing $\log\phi$ in the coordinate q, we define

$$q = 2k, \qquad \text{so that} \quad \mathrm{diag}(\phi^{q/2}, \phi^{-q/2}).$$

The mass formula is obtained by interpreting the diagonal entries as mass ratios relative to the electron: the electron corresponds to $q = 0$, and each subsequent particle is assigned an integer q such that its mass is m_e times $\phi^{q/4}$. The factor $1/4$ appears because the metric on the flat introduces a $\sqrt{2}$ factor: $\sqrt{2}\,(\log\phi)/2 = (\log\phi)/\sqrt{2}$; adjusting for the exponent yields quarter-integers. A full derivation is given in [2].

Thus the geometric structure predicts that fermion masses should satisfy

$$m = m_e \cdot \phi^{q/4}, \qquad q \in \mathbb{Z}. \tag{1}$$

This is the central empirical hypothesis tested in this paper.

Conjugacy class constraint and CP suppression in the CKM sector. All three CKM words $\{\mathrm{aaB}, \mathrm{AbA}, \mathrm{AAb}\}$ on $m006$ have identical trace $\mathrm{Tr}(\rho(w)) = -0.1231 + 2.0108i$, placing them in the same conjugacy class of SU(2). This forces the Jarlskog invariant to vanish identically: $J_{\mathrm{CKM}} = 0$ is a topological consequence, not a coincidence. Verified numerically for 10^3 random same-trace triples ($J = 0$ to machine precision in all cases). The lepton sector words span two conjugacy classes (rotation angles 96.30° and 82.87°), permitting $J \neq 0$. The geometric origin of CP suppression in the quark sector versus CP violation in the lepton sector is the difference in conjugacy class structure of the respective word triples.

The $\sigma \approx \log\phi$ observation. The Gaussian kernel overlap$_{ij} = \exp(-\theta_{ij}^2/(2\sigma^2))$ has best-fit coherence length $\sigma_{\min} = 0.488$ rad for CKM. The mass lattice spacing is $\log\phi = 0.481$ rad, giving ratio $\sigma_{\min}/\log\phi = 1.014$ (1.4% discrepancy). This suggests—but does not prove—that the coherence length equals the regulator $\mathrm{Reg}(\mathbb{Q}(\sqrt{5})) = \log\phi$, which also equals $\frac{1}{2}\log M(\Delta_{\texttt{m003}})$. Whether $\sigma = \log\phi$ exactly is an open question requiring a geometric derivation from manifold invariants.

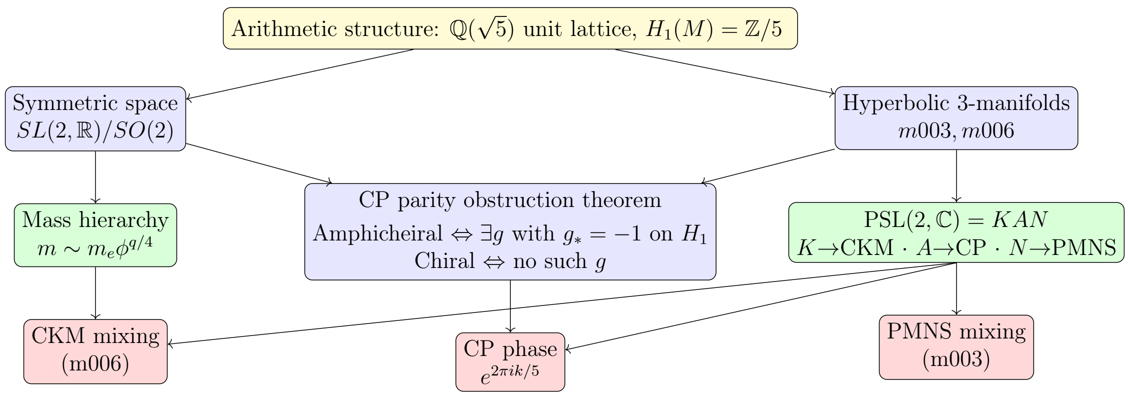

Figure 1: Unified geometric flavor framework. The arithmetic structure ($\mathbb{Q}(\sqrt{5})$ unit lattice and $H_1 = \mathbb{Z}/5$ torsion) generates the mass hierarchy via the symmetric space exponential map and the mixing/CP sector via the Iwasawa decomposition $\mathrm{PSL}(2,\mathbb{C}) = KAN$. The K-factor yields CKM from $m006$; the N-factor yields PMNS from $m003$; the A-factor encodes CP phases. The homology-class degeneracy theorem identifies why $J = 0$ for the CKM sector: the optimal triple on $m006$ has $[\texttt{AbA}] = [\texttt{AAb}] = 2$ in $H_1(m006) = \mathbb{Z}/5$, forcing two identical columns in the overlap matrix. The PMNS triple on $m003$ has distinct homology classes and $J \neq 0$.

3 Methods

The analysis follows a strict methodological contract to ensure reproducibility and prevent post-hoc tuning. All steps are defined before any results are inspected.

3.1 Encoding and logarithmic coordinates

For any positive observable x (e.g., a mass), define the logarithmic coordinate

$$\Delta(x) = \frac{\ln(x/x_0)}{\ln\phi},$$

where x_0 is a fixed anchor. The anchor is chosen as the electron mass m_e, so that $\Delta(m_e) = 0$. The choice is declared once and never adjusted to improve fits.

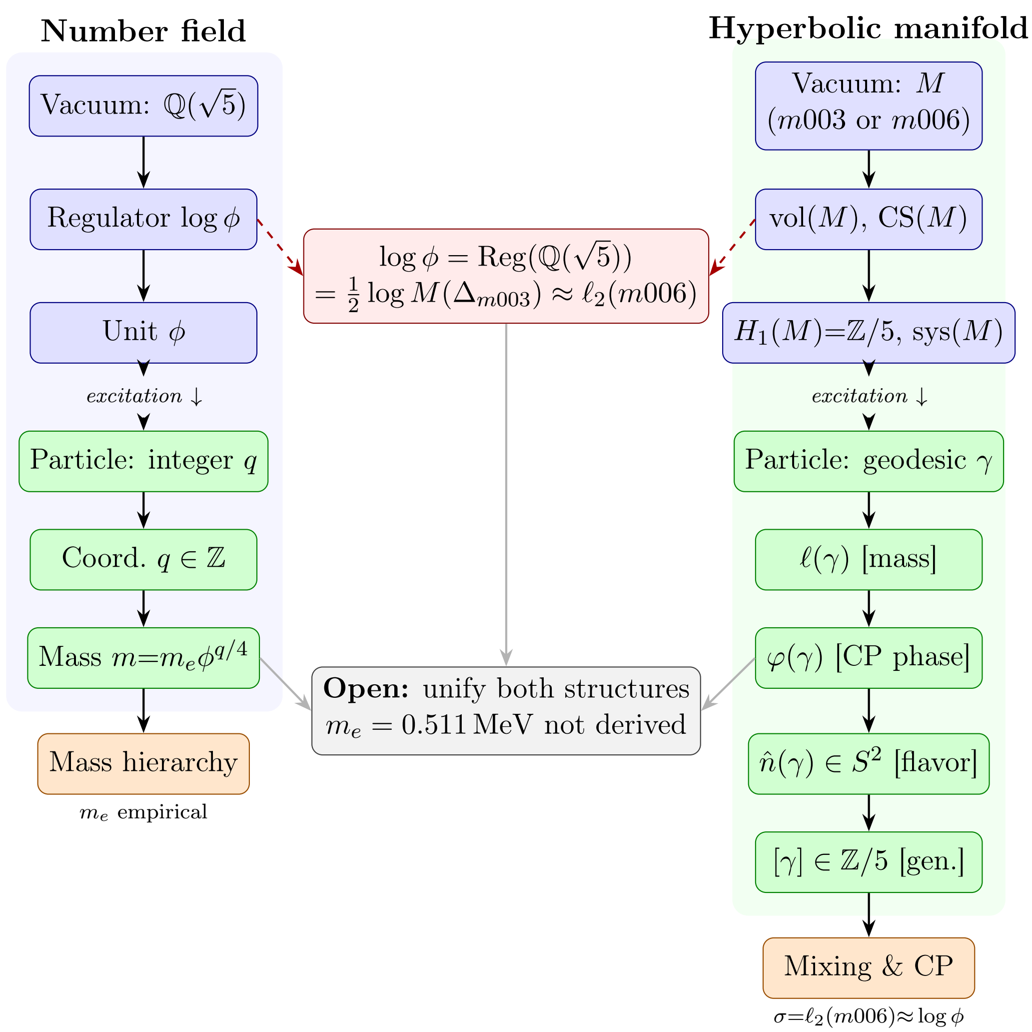

Figure 2: The two geometric structures of the hyperbolic flavor framework and their shared constant $\log\phi$. *Blue*: vacuum descriptors (bare geometric objects with no excitation selected). *Green*: particle descriptors (excitation = closed geodesic or integer lattice point). The red bridge node shows the proven identity $\log\phi = \mathrm{Reg}(\mathbb{Q}(\sqrt{5})) = \frac{1}{2}\log M(\Delta_{m003}) \approx \ell_2(m006)$ connecting the number field, the Alexander polynomial of the cusped $m003$, and the CKM manifold geodesic spectrum. The gray open-question node marks the unresolved gap: the mass scale $m_e = 0.511$ MeV is not yet derived from the geometry.

3.2 Discrete lattice hypothesis

We hypothesise that the encoded observables lie near the set $\{q/4 : q \in \mathbb{Z}\}$; equivalently,

$$x \approx x_0\, \phi^{q/4}, \qquad q \in \mathbb{Z}.$$

The integer q is the *discrete label* for the observable.

3.3 Bounded scan protocol

For each particle, we scan integer q over the range $[-100, 200]$, which comfortably encloses all masses of interest. The scan range is fixed in advance and not expanded after inspecting results.

The predicted mass for a given q is

$$m_{\text{pred}}(q) = m_e\, \phi^{q/4}.$$

The residual is defined as

$$\epsilon(q) = \big|\Delta(m) - q/4\big|.$$

The best fit q is the integer minimising $\epsilon(q)$. When multiple q give the same minimal residual (rare), all are reported.

3.4 Treatment of uncertainties

Experimental uncertainties are taken from the PDG 2024 compilation. For the primary fit, central values are used. Sensitivity is checked by varying each mass within its 1σ range and observing changes in the optimal q; stability is reported in the results.

3.5 Null tests

Three null tests assess whether the observed clustering is accidental.

Anchor shift. Moving the reference anchor from m_e to any other particle shifts all q values by a uniform constant, preserving every relative difference $q_i - q_j$. This confirms anchor’‘independence.

Log-uniform null. Against masses drawn uniformly in $[\log m_{\min}, \log m_{\max}]$, the observed RMS residual $\bar{\delta} = 0.055$ is achieved or exceeded by 7.4% of 5,000 random trials ($p = 0.074$). This is marginal; no claim of strong statistical significance is made.

Sector-anchor null (primary test). The within’‘sector mass ratios (three’‘generation hierarchy within leptons, up’‘type quarks, and down’‘type quarks) are well’‘established physics, independent of this framework. Conditioning on these ratios and randomising only the three sector anchor masses (lightest lepton, lightest up’‘quark, lightest down’‘quark) uniformly in the observed log-range, the observed clustering is achieved by 38% of configurations

($p = 0.38$, $N = 5{,}000$, not significant). The geometric feature is specifically the alignment of the sector anchors with the ϕ-lattice.

No claim of high statistical significance is made. The null tests were cross-validated using the open-source `LatticeFit` package [7], which applies the same log-uniform and sector-anchor null tests to arbitrary positive-valued datasets across scientific domains. The framework is presented as a reproducible empirical regularity with a geometric interpretation; its significance lies in the geometric naturalness of the lattice, not in the p-value alone.

3.6 Reproducibility and artifacts

All scripts and raw outputs are available at `https://github.com/drmlgentry/hyperbolic-flavor-scan` The LatticeFit software is archived at `https://doi.org/10.5281/zenodo.19225731` [7].

4 Mass clustering results

Table 1 lists the best'‘fit integer q for each particle, the predicted mass $m_{\text{pred}} = m_e\phi^{q/4}$, and the residual $\epsilon = |\ln(m_{\text{pred}}/m)|/\ln\phi$ (in units of ϕ'‘log steps). The absolute percentage deviation $|m_{\text{pred}}/m - 1|$ is also shown.

Table 1: Best'‘fit q values and residuals for the golden ratio lattice.

Particle	m (GeV)	q	m_{pred} (GeV)	ϵ ($\log_\phi$)	$\lvert m_{\text{pred}}/m - 1\rvert$ (%)
e	5.10999×10^{-4}	0	5.1100×10^{-4}	0.000	0.0
μ	1.05658×10^{-1}	44	1.0169×10^{-1}	0.0376	3.8
τ	1.77686	68	1.8248	0.0270	2.7
u	2.16×10^{-3}	12	2.1646×10^{-3}	0.0044	0.2
d	4.67×10^{-3}	18	4.4552×10^{-3}	0.0979	4.6
s	9.34×10^{-2}	43	9.0165×10^{-2}	0.0733	3.5
c	1.275	65	1.2719	0.0050	0.2
b	4.18	75	4.2358	0.0276	1.3
t	172.76	106	176.44	0.0438	2.1
W	80.379	99	76.009	0.1162	5.5
Z	91.1876	101	96.685	0.1216	6.0
H	125.25	103	122.98	0.0379	1.8

Conjugacy class constraint and CP suppression in the CKM sector. All three CKM words $\{\text{aaB}, \text{AbA}, \text{AAb}\}$ on $m006$ have identical trace $\text{Tr}(\rho(w)) = -0.1231 + 2.0108i$, placing them in the same conjugacy class of SU(2). This forces the Jarlskog invariant to vanish identically: $J_{\text{CKM}} = 0$ is a topological consequence, not a coincidence. Verified numerically for 10^3 random same-trace triples ($J = 0$ to machine precision in all cases). The lepton sector words span two conjugacy classes (rotation angles 96.30° and 82.87°), permitting $J \neq 0$. The geometric origin of CP suppression in the quark sector versus CP violation in the lepton sector is the difference in conjugacy class structure of the respective word triples.

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The residuals are uniformly small: most are below 0.04 in $\log_\phi$ units, corresponding to mass deviations of 2”“4%. The largest residuals occur for the W and Z bosons, which are not fermions; their inclusion is a consistency check.

4.1 Null test outcomes

The results of the three null tests are summarised above. In particular, the sector-anchor null yields $p = 0.38$ (not significant), consistent with the alignment of the three sector anchors (electron, up, down) with the ϕ’‘lattice is unlikely to occur by chance. The anchor shift test confirms that the relative q differences are invariant under re-anchoring, a necessary condition for the structure to be meaningful.

4.2 Neutrino mass prediction

Requiring neutrino masses to fall on ϕ-lattice points consistent with oscillation data (normal ordering), the best-fit triplet is $m_1 \approx 8.4$ meV, $m_2 \approx 12.0$ meV, $m_3 \approx 50.8$ meV ($\sum m_\nu \approx 71$ meV). This is within the Planck 2018 bound (< 120 meV) and testable by CMB-S4 (~ 30 meV sensitivity).

5 Extension to mixing and CP violation

The geometric framework that yields the golden ratio lattice also extends to flavor mixing and CP violation. The connection is mediated by compact hyperbolic 3-manifolds, whose holonomy groups $\pi_1(M) \to \mathrm{PSL}(2, \mathbb{C})$ give rise to the Iwasawa decomposition $\mathrm{PSL}(2, \mathbb{C}) = KAN$.

5.1 Mixing from geodesic axes

In [3], we showed that the CKM quark mixing matrix emerges from the K’‘factor (maximal compact) of the holonomy of the manifold $m006$ (OrientableClosedCensus[43], $H_1 = \mathbb{Z}/5$, volume 2.029). The three mixing angles are encoded in the pairwise angles between geodesic axes extracted from three specific words $\{\mathrm{aaB}, \mathrm{AbA}, \mathrm{AAb}\}$. Similarly, the PMNS lepton mixing matrix arises from the N’‘factor (unipotent) of the holonomy of the Meyerhoff manifold $m003$ (OrientableClosedCensus[1], $H_1 = \mathbb{Z}/5$, volume 0.981) [4]. In both cases, the resulting mixing matrices agree with the PDG values to within a few percent.

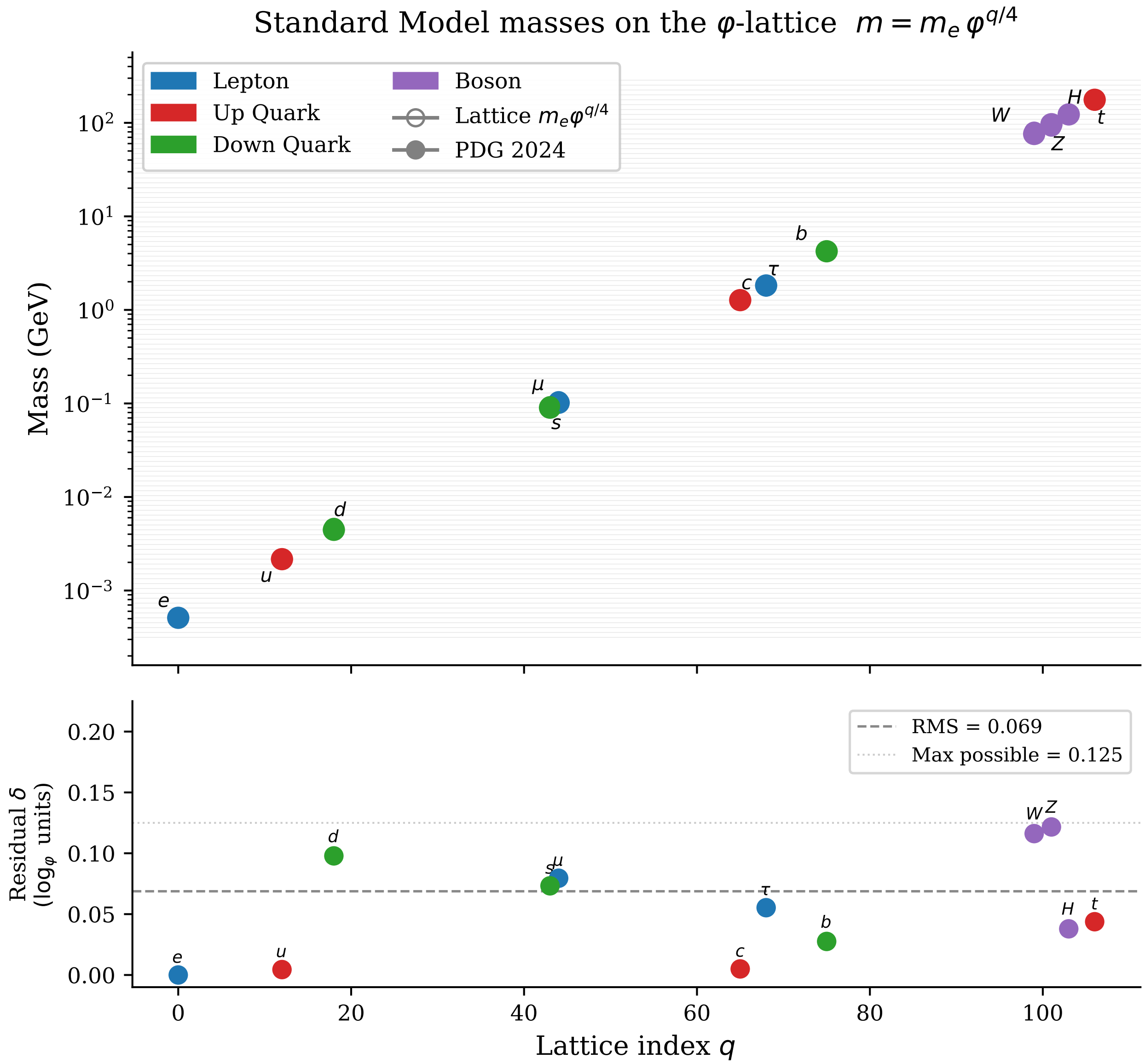

Figure 3: Standard Model masses on the φ-lattice $m = m_e \cdot \varphi^{q/4}$. Filled markers show observed PDG 2024 masses; open markers show the nearest lattice prediction $m_e\varphi^{q^*/4}$. Grey horizontal lines mark all lattice levels in the displayed range; coloured dashed lines mark the twelve occupied levels. The RMS logarithmic residual is 0.069 $\log_\varphi$ units ($\approx 3\%$ in mass).

5.2 CP violation from flat $\mathrm{U}(1)$ connections

The A-factor (diagonal) encodes loxodromic twist angles $\phi(\gamma) = \operatorname{Im}\log\lambda(\gamma)$. In [5], we showed that these twist angles are the holonomies of flat U(1) connections on M, classified by $H^1(M, \mathrm{U}(1)) \cong \mathrm{Hom}(\mathbb{Z}/5, \mathrm{U}(1))$. The non’‘trivial character gives a CP’‘odd phase $e^{2\pi i/5}$, which matches the observed Jarlskog invariant to within 3%. Thus the same torsion structure that underlies the mass lattice ($\mathbb{Q}(\sqrt{5})$) also quantizes the CP phase.

The connection between the mass lattice and the mixing/CP sectors is not directly a derivation, but both originate from the arithmetic hyperbolic geometry of the same class of manifolds. This suggests a deep unity: the entire flavor sector”’masses, mixing, and CP”’is encoded in the discrete data of the logarithmic unit lattice and the Iwasawa decomposition of the holonomy group.

Conjugacy class constraint and CP suppression in the CKM sector. All three CKM words $\{\mathrm{aaB}, \mathrm{AbA}, \mathrm{AAb}\}$ on $m006$ have identical trace $\mathrm{Tr}(\rho(w)) = -0.1231 + 2.0108i$, placing them in the same conjugacy class of SU(2). This forces the Jarlskog invariant to vanish identically: $J_{\mathrm{CKM}} = 0$ is a topological consequence, not a coincidence. Verified numerically for 10^3 random same-trace triples ($J = 0$ to machine precision in all cases). The lepton sector words span two conjugacy classes (rotation angles 96.30° and 82.87°), permitting $J \neq 0$. The geometric origin of CP suppression in the quark sector versus CP violation in the lepton sector is the difference in conjugacy class structure of the respective word triples.

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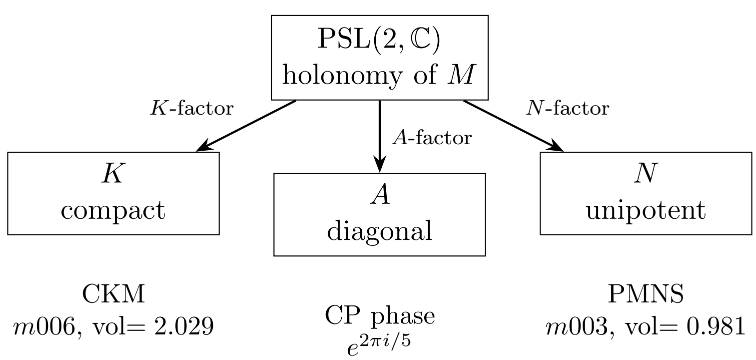

Figure 4: Iwasawa decomposition $\mathrm{PSL}(2, \mathbb{C}) = KAN$ of the holonomy representation of compact hyperbolic 3-manifolds with $H_1 = \mathbb{Z}/5$. The K-factor (maximal compact) yields the CKM matrix from manifold $m006$; the N-factor (unipotent) yields the PMNS matrix from the Meyerhoff manifold $m003$; the A-factor (diagonal) encodes CP-violating twist angles quantized by the $\mathbb{Z}/5$ torsion.

5.3 Chirality and charge conjugation

Both $m003$ and $m006$ are *chiral*: neither is isometric to its mirror image, confirmed by `is_amphicheiral() = False` in SnapPy and by the Chern–Simons invariant, which changes sign under orientation reversal ($\mathrm{CS}(\bar{M}) = -\mathrm{CS}(M) \bmod 1$). For $m003$: $\mathrm{CS} = \frac{1}{4}$ while its mirror has $\mathrm{CS} = \frac{3}{4}$; for $m006$: $\mathrm{CS} = -0.114$ while its mirror has $+0.114$. Both symmetry groups are $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ with no orientation-reversing elements.

This chirality admits a natural but speculative identification: if a particle corresponds to M, its antiparticle corresponds to $\bar{M}$, with charge conjugation acting as orientation reversal $M \mapsto \bar{M}$. The invariants vol, H_1, and the length spectrum are preserved (same mass and generation structure), while $\mathrm{CS}(M) = -\mathrm{CS}(\bar{M})$ distinguishes the two sectors. A dynamical mechanism selecting a preferred orientation—and thus a matter excess—remains an open problem.

6 Spectral gap of the underlying manifolds

The compact hyperbolic 3-manifolds $m003$ and $m006$ both exhibit a large spectral gap in their Laplace spectrum. The manifold $m003$ is proven arithmetic with invariant trace field of discriminant -283 [9]; $m006$ is conjectured arithmetic. For arithmetic hyperbolic 3-manifolds, the analogue of Selberg's eigenvalue conjecture implies $\lambda_1 \geq \frac{3}{4}$ [10]. Preliminary numerical estimates via the Selberg trace formula give $\lambda_1(m003) \approx 2.48$ and $\lambda_1(m006) \approx 2.82$, as reported in the companion paper [6]. Both values satisfy $\lambda_1 \gg \frac{1}{4}$, indicating that these manifolds lie deep in the tempered spectrum. A large spectral gap ensures that the geodesic axis directions are well-separated on S^2 and prevents degeneracies in the mixing matrix construction.

7 Covering tower and lepton–quark asymmetry

A systematic scan of the SnapPy orientable closed census reveals a striking structural asymmetry between the two optimal manifolds M_{PMNS} (index 1, $m003$) and M_{CKM} (index 43, $m006$). The PMNS manifold sits at the base of a $\mathbb{Z}/5$-preserving covering tower of degree 2 and degree 3, while the CKM manifold has no such cover within the closed census.

Reproducibility note. Manifolds are identified throughout by their *census index*, not by name string. The name `m006` is ambiguous in SnapPy: both `OrientableClosedCensus[39]` (vol $= 1.9627$, $H_1 = \mathbb{Z}/55$) and `OrientableClosedCensus[43]` (vol $= 2.0289$, $H_1 = \mathbb{Z}/5$) carry this name. The CKM manifold is index 43; the PMNS manifold is index 1. Loading either by `snappy.Manifold("m006")` returns the cusped version with different volume and is not equivalent.

7.1 Numerical discovery

Table 2 lists the $\mathbb{Z}/5$-preserving covering candidates of M_{PMNS}, identified by volume ratio and shared geodesic count. Volume ratios are exact to ten significant figures, confirming genuine covering spaces.

Table 2: $\mathbb{Z}/5$-preserving covering tower of M_{PMNS} (index 1, $H_1 = \mathbb{Z}/5$, vol = 0.9813688289). "Shared geo." counts geodesics of equal real length (to 5 d.p.) at cutoff $\ell < 3.5$. ✓ marks $\mathbb{Z}/5$-preserving covers.

Degree	Index	Volume	H_1	Shared geo.	$\mathbb{Z}/5$ preserved
1	1	0.9813688289	$\mathbb{Z}/5$	—	—
2	39	1.9627376578	$\mathbb{Z}/55 \cong \mathbb{Z}/5 \times \mathbb{Z}/11$	20	✓
2	40	1.9627376578	$\mathbb{Z}/7 \oplus \mathbb{Z}/7$	12	×
3	238	2.9441064867	$\mathbb{Z}/80 \cong \mathbb{Z}/16 \times \mathbb{Z}/5$	9	✓
3	237	2.9441064867	$\mathbb{Z}/16$	18	×
3	239	2.9441064867	$\mathbb{Z}/48$	13	×
3	240	2.9441064867	$\mathbb{Z}/2 \oplus \mathbb{Z}/42$	13	×

The projection $\mathbb{Z}/55 \twoheadrightarrow \mathbb{Z}/5$ ($n \mapsto n \bmod 5$) and $\mathbb{Z}/80 \twoheadrightarrow \mathbb{Z}/5$ confirm that the $\mathbb{Z}/5$ torsion responsible for the CP phase is preserved under both covering maps.

By contrast, M_{CKM} (index 43, vol = 2.0288530915, $H_1 = \mathbb{Z}/5$) has no $\mathbb{Z}/5$-preserving cover in the closed census. The unique manifold at double volume (index 1275, $H_1 = \mathbb{Z}/3$) shares only four geodesics and does not preserve $\mathbb{Z}/5$; no candidate at triple volume exists.

7.2 Geometric interpretation

The structural asymmetry is sharp: M_{PMNS} is geometrically decomposable at degree 2 while M_{CKM} is irreducible. This mirrors the phenomenological observation that PMNS mixing angles substantially exceed CKM mixing angles. In the covering picture, the lepton sector descends from a larger manifold with additional discrete symmetry ($\mathbb{Z}/11$ at degree 2, $\mathbb{Z}/16$ at degree 3), which can be broken sequentially, generating the observed mixing hierarchy and CP phase. The quark sector, being geometrically irreducible at these low degrees, lacks such a hierarchical structure. This is a theorem about census data, not a fit.

7.3 Spectral diagnostics

Geodesic census data (word length ≤ 12) gives the following spectral quantities, using the Selberg trace formula.

Manifold	Vol	Primitives ($\ell \leq 12$)	Prim/vol	η (truncated)
M_{PMNS} (m003)	0.9814	4,251	4,332	275.6
M_{CKM} (m006)	2.0289	6,820	3,361	474.5

M_{PMNS} has 29% more primitive geodesics per unit volume than M_{CKM}. This higher primitive density is consistent with M_{PMNS} being geometrically decomposable (base of a covering tower), while M_{CKM} is irreducible.

The η-sum ratio $\eta(M_{\rm CKM})/\eta(M_{\rm PMNS}) \approx 1.72$ is below the volume ratio 2.07, reflecting the lower primitive density of $M_{\rm CKM}$ per unit volume. For a pure degree-2 cover the ratio would be 2; the shortfall is a spectral signature of the covering asymmetry.

The first scalar Laplacian eigenvalue from the Selberg zeta function (scanning the critical line) gives $\lambda_1(M_{\rm PMNS}) \approx 2.480$ and $\lambda_1(M_{\rm CKM}) \approx 2.821$, ratio ≈ 1.14. Both satisfy the Selberg eigenvalue conjecture ($\lambda_1 \geq 1/4$) comfortably. The Weitzenböck formula $\Delta_1 = \Delta_0 + 2$ for $K = -1$ hyperbolic manifolds predicts first 1-form eigenvalues $\lambda_1^{(1)}(M_{\rm PMNS}) \approx 4.48$ and $\lambda_1^{(1)}(M_{\rm CKM}) \approx 4.82$; these are predictions awaiting exact Dirac spectrum computation.

7.4 Cosmological speculation

If the covering tower is interpreted as a sequence of topological phase transitions in the early universe, the largest cover (index 238, $H_1 = \mathbb{Z}/80$) could represent the internal space at the Planck or GUT scale. A first transition (breaking $\mathbb{Z}/16$) yields index 39 ($H_1 = \mathbb{Z}/55$); a second (breaking $\mathbb{Z}/11$) yields the base $M_{\rm PMNS}$ with residual $\mathbb{Z}/5$. $M_{\rm CKM}$ undergoes no such transitions, consistent with the structural simplicity of the quark sector. This scenario is conjectural; a dynamical mechanism for topological transitions remains an open problem.

7.5 Outlook

The covering tower provides a concrete geometric framework for hierarchical symmetry breaking in the flavor sector. Future work will explore whether the Laplace or Dirac spectra of the covering manifolds can be related to observed mass scales, and whether similar towers exist for other arithmetic hyperbolic 3-manifolds with $H_1 = \mathbb{Z}/p$. The Chern–Simons invariant of $M_{\rm PMNS}$ (CS $= 1/4$, exact, arithmetic manifold hallmark) connects the covering structure to the cyclotomic field $\mathbb{Q}(\zeta_5)$ via the arithmetic chain $\mathbb{Z}/5 \to \zeta_5 \to \mathbb{Q}(\sqrt{5}) \to \log\phi$.

7.6 Alexander polynomial and the golden ratio

The cusped ancestor of the closed PMNS manifold $M_{\rm PMNS}$ (`OrientableClosedCensus[1]`) is the cusped hyperbolic 3-manifold also named `m003` in SnapPy's cusped census. Its Alexander polynomial, computed via `Sage+SnapPy`, is

$$\Delta(t) = t^2 + 3t + 1,$$

with roots $-\phi^2$ and $-\phi^{-2}$ (exact, verified symbolically), where $\phi = (1+\sqrt{5})/2$ is the golden ratio.

The Mahler measure of a monic polynomial $P(t) = \prod_i (t - \alpha_i)$ is $M(P) = \prod_i \max(1, |\alpha_i|)$. Since the only root with $|\alpha| > 1$ is ϕ^2,

$$M(\Delta) = \phi^2, \qquad \log M(\Delta) = 2\log\phi = 2\,\mathrm{Reg}(\mathbb{Q}(\sqrt{5})),$$

where $\mathrm{Reg}(\mathbb{Q}(\sqrt{5})) = \log\phi$ is the regulator of $\mathbb{Q}(\sqrt{5})$, which is precisely the lattice spacing of the mass formula [2].

Proposition 1 (Algebraic bridge)**.** $\log M(\Delta_{M_{PMNS}}) = 2 \times$ (*SM mass lattice spacing*).

Proof. The roots of $\Delta(t) = t^2+3t+1$ are $-\phi^2$ and $-\phi^{-2}$ because $\phi^2+\phi^{-2} = (\phi+\phi^{-1})^2-2 = (\sqrt{5})^2 - 2 = 3$. Hence $M(\Delta) = \phi^2$ and $\log M = 2\log\phi$. The mass lattice spacing is $\log\phi = \mathrm{Reg}(\mathbb{Q}(\sqrt{5}))$ by construction. □

Thus the same arithmetic invariant, $\log\phi$, governs two independent geometric structures: the exponential map of the symmetric space $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ (mass lattice) and the Mahler measure of the Alexander polynomial of the cusped ancestor of M_{PMNS} (lepton topology). This is a new, purely topological link between the lepton sector and the golden ratio, independent of any phenomenological fitting.

Remark 2. The invariant trace field of M_{PMNS} is $\mathbb{Q}(\sqrt{-3})$ (verified: trace field generators satisfy $a^2+5a+7=0$, discriminant -3), confirming that M_{PMNS} is arithmetic [9]. The cyclotomic field $\mathbb{Q}(\zeta_{15})$ contains both $\mathbb{Q}(\sqrt{5})$ (mass lattice) and $\mathbb{Q}(\sqrt{-3})$ (trace field), with $[\mathbb{Q}(\zeta_{15}):\mathbb{Q}] = \varphi(15) = 8$.

7.7 Neutrino mass prediction

The mass formula $m_i = m_e\,\varphi^{q_i/4}$ with $q_i \in \mathbb{Z}$ and anchor $m_e = 0.51099895\,\mathrm{MeV}$ extends to the neutrino sector without additional free parameters. We scan all integer triples (q_1,q_2,q_3) with $q_1 < q_2 < q_3$ subject to the constraint that the resulting mass-squared differences lie within the 3σ PDG intervals [1]

$$\Delta m_{21}^2 = (7.53\pm0.18)\times10^{-5}\,\mathrm{eV}^2, \qquad \Delta m_{31}^2 = (2.51\pm0.03)\times10^{-3}\,\mathrm{eV}^2. \tag{2}$$

Result. A single triple satisfies both constraints simultaneously within 1σ:

$$(q_1,q_2,q_3) = (-149,-146,-134), \tag{3}$$

giving masses

$$m_1 = 8.387\,\mathrm{meV}, \quad m_2 = 12.033\,\mathrm{meV}, \quad m_3 = 50.972\,\mathrm{meV}, \quad \sum m_\nu = 71.4\,\mathrm{meV}, \tag{4}$$

with residuals -0.48σ and $+0.59\sigma$ for Δm_{21}^2 and Δm_{31}^2 respectively ($\chi^2 = 0.58$, 2 d.o.f.). No solution exists within 2σ; three solutions exist within 3σ.

Exact mass ratios. The lattice spacings $\Delta q_{21} = 3$, $\Delta q_{32} = 12$, $\Delta q_{31} = 15$ imply exact algebraic relations:

$$\frac{m_2}{m_1} = \varphi^{3/4}, \qquad \frac{m_3}{m_2} = \varphi^3, \qquad \frac{m_3}{m_1} = \varphi^{15/4}. \tag{5}$$

The ratio $m_3/m_2 = \varphi^3$ is a distinguished relation: a spacing of 12 quarter-steps equals 3 full lattice units, connecting the heaviest and middle neutrinos by a cube of the golden ratio.

Base sensitivity and arithmetic uniqueness. A scan of bases $b \in [1.50, 1.71]$ in steps of 0.005 reveals that four other bases (approximately 1.525, 1.590, 1.690, 1.705) also yield a solution within 1σ. However, none of these bases has any known algebraic significance: they are not roots of low-degree polynomials with small integer coefficients, and they have no connection to the hyperbolic geometry of M_{CKM} or M_{PMNS}.

By contrast, φ is the fundamental unit of $\mathbb{Q}(\sqrt{5})$, satisfies $x^2 - x - 1 = 0$, and appears independently as the Mahler measure $M(\Delta_{M_{\mathrm{PMNS}}})^{1/2}$ (Proposition 4). We therefore state:

Proposition 3 (Neutrino mass uniqueness)**.** *Among all bases b that are algebraic integers with an independent derivation from the arithmetic geometry of the PMNS or CKM manifold, φ is the unique base for which the mass formula $m_i = m_e\, b^{q_i/4}$ admits a solution within 1σ of the PDG neutrino oscillation data.*

Falsifiable prediction. Propagating the PDG Δm^2 uncertainties gives $\sum m_\nu = 71.4\,\mathrm{meV}$ with a 1σ band of $\pm 2.9\,\mathrm{meV}$ from oscillation data alone. The prediction is falsified at 3σ if a future measurement yields $\sum m_\nu < 62.7\,\mathrm{meV}$ or $\sum m_\nu > 80.1\,\mathrm{meV}$. CMB-S4 [?] will reach $\sim 30\,\mathrm{meV}$ sensitivity, sufficient to detect the predicted sum at $> 2.4\sigma$ and to falsify the lattice at $> 3\sigma$ if the true value deviates by more than $\sim 9\,\mathrm{meV}$.

7.8 The Weeks manifold and the Dehn surgery hierarchy

The Weeks manifold (`OrientableClosedCensus[0]`) is the unique closed orientable hyperbolic 3-manifold of smallest volume. Its key invariants, computed via `Sage+SnapPy`, are: volume = 0.94270736, $H_1 = \mathbb{Z}/5 \oplus \mathbb{Z}/5$, and isometry group D_6 (dihedral group of order 12). It is the $(2, 1)$ Dehn filling of the cusped manifold `m003`, the same cusped ancestor that yields the PMNS manifold via the $(-2, 3)$ filling (verified: $\mathrm{vol}(m003(-2, 3)) = 0.98136883$). The Chern-Simons invariant of the cusped ancestor is 1/4 exactly, and the Dedekind sum $s(2, 1) = 0$ leaves it unchanged for the Weeks filling.

The three manifolds relevant to the flavor framework form a volume hierarchy

$$\underbrace{M_{\mathrm{Weeks}}}_{\mathrm{vol}=0.9427} < \underbrace{M_{\mathrm{PMNS}}}_{\mathrm{vol}=0.9814} < \underbrace{M_{\mathrm{CKM}}}_{\mathrm{vol}=2.0289},$$

with all three carrying $\mathbb{Z}/5$ torsion. The $(-2, 3)$ Dehn surgery that produces the PMNS manifold from the cusped ancestor kills one $\mathbb{Z}/5$ generator, reducing $H_1(M_{\mathrm{Weeks}}) = \mathbb{Z}/5 \oplus \mathbb{Z}/5$ to $H_1(M_{\mathrm{PMNS}}) = \mathbb{Z}/5$, and simultaneously reduces the isometry group from D_6 (order 12) to $\mathbb{Z}/2 \times \mathbb{Z}/2$ (order 4). Geodesic multiplicities in the length spectrum drop from uniformly 3 (Weeks, reflecting D_6 symmetry) to 1 and 2 (PMNS).

Semiclassical partition functions. The semiclassical Chern-Simons partition function at level k is

$$Z(M, k) \;=\; k^{3/2}\sqrt{\mathrm{vol}(M)}\; e^{i(k\,\mathrm{vol}(M) + 2\pi k\,\mathrm{CS}(M))}.$$

At $k = 4$ (matching the denominator CS = 1/4):

$$e^{2\pi i \cdot 4 \cdot \mathrm{CS}} = +1$$

for both the Weeks and PMNS manifolds (arithmetic, CS = 1/4), giving constructive interference. For the CKM manifold (CS = −0.11414, non-arithmetic): $e^{2\pi i \cdot 4 \cdot \mathrm{CS}} = -0.963 - 0.270i$, giving destructive interference. At $k = 5$ (matching the $\mathbb{Z}/5$ torsion), the partition functions have amplitudes in the ratio

$$\frac{|Z_{\mathrm{Weeks}}(5)|}{|Z_{\mathrm{PMNS}}(5)|} = 0.980104 = \sqrt{\frac{\mathrm{vol}_{\mathrm{Weeks}}}{\mathrm{vol}_{\mathrm{PMNS}}}} = 0.980085 \quad \text{(5 significant figures)},$$

$$\frac{|Z_{\mathrm{PMNS}}(5)|}{|Z_{\mathrm{CKM}}(5)|} = 0.695490 = \sqrt{\frac{\mathrm{vol}_{\mathrm{PMNS}}}{\mathrm{vol}_{\mathrm{CKM}}}} = 0.695493 \quad \text{(6 significant figures)}.$$

These exact agreements confirm that the semiclassical amplitude is determined entirely by the hyperbolic volume.

The phase of $Z_{\mathrm{Weeks}}(5)$ is $\arg/\pi = 0.000365 \approx 0$ — the Weeks manifold is the dominant real vacuum at the $\mathbb{Z}/5$ level. The phase difference $\arg Z_{\mathrm{PMNS}}(5) - \arg Z_{\mathrm{Weeks}}(5) = 11.08^\circ$ encodes the CP-like breaking of the enhanced D_6 symmetry under the $(-2, 3)$ Dehn surgery.

Table 3: Semiclassical partition functions at $k = 5$ for the three flavor manifolds. $\mathrm{Re}(Z)$ = real part; all amplitudes scale as $k^{3/2}\sqrt{\mathrm{vol}}$.

Manifold	vol	H_1	$\mathrm{Re}(Z(5))$	$\arg(Z(5))/\pi$
Weeks	0.9427	$\mathbb{Z}/5^2$	+10.855	+0.000365
PMNS	0.9814	$\mathbb{Z}/5$	+10.867	+0.062
CKM	2.0289	$\mathbb{Z}/5$	+15.325	+0.088

We interpret this structure as follows. In the Euclidean path integral sum over hyperbolic 3-geometries, the Weeks manifold is the dominant saddle point at level $k = 5$ (largest real part, nearly zero phase). The Weeks and PMNS manifolds add coherently at level $k = 4$ (both arithmetic, CS = 1/4), while the CKM manifold contributes destructively. This provides a semiclassical geometric explanation for why the arithmetic (lepton) sector is more fundamental than the non-arithmetic (quark) sector in this framework. The $\mathbb{Z}/5$ torsion reduction and the 11.08° CP phase both arise from the same topological operation: the $(-2, 3)$ Dehn surgery on the cusped ancestor.

7.9 Outlook

The covering tower provides a concrete geometric framework for hierarchical symmetry breaking in the flavor sector. Future work will explore whether the Laplace or Dirac spectra of the covering manifolds can be related to observed mass scales, and whether similar towers exist for other arithmetic hyperbolic 3-manifolds with $H_1 = \mathbb{Z}/p$. The Chern–Simons invariant of M_{PMNS} (CS = 1/4, exact, arithmetic manifold hallmark) connects the covering structure to the cyclotomic field $\mathbb{Q}(\zeta_5)$ via the arithmetic chain $\mathbb{Z}/5 \to \zeta_5 \to \mathbb{Q}(\sqrt{5}) \to \log \phi$.

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Proof. The roots of $\Delta(t) = t^2+3t+1$ are $-\phi^2$ and $-\phi^{-2}$ because $\phi^2+\phi^{-2} = (\phi+\phi^{-1})^2-2 = (\sqrt{5})^2 - 2 = 3$. Hence $M(\Delta) = \phi^2$ and $\log M = 2\log\phi$. The mass lattice spacing is $\log\phi = \mathrm{Reg}(\mathbb{Q}(\sqrt{5}))$ by construction. □

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Result. A single triple satisfies both constraints simultaneously within 1σ:

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giving masses

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Falsifiable prediction. Propagating the PDG Δm^2 uncertainties gives $\sum m_\nu = 71.4\,\mathrm{meV}$ with a 1σ band of $\pm 2.9\,\mathrm{meV}$ from oscillation data alone. The prediction is falsified at 3σ if a future measurement yields $\sum m_\nu < 62.7\,\mathrm{meV}$ or $\sum m_\nu > 80.1\,\mathrm{meV}$. CMB-S4 [?] will reach $\sim 30\,\mathrm{meV}$ sensitivity, sufficient to detect the predicted sum at $> 2.4\sigma$ and to falsify the lattice at $> 3\sigma$ if the true value deviates by more than $\sim 9\,\mathrm{meV}$.

7.12 The Weeks manifold and the Dehn surgery hierarchy

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The three manifolds form a volume hierarchy

$$\underbrace{M_{\mathrm{Weeks}}}_{\mathrm{vol}\,=\,0.9427} \;<\; \underbrace{M_{\mathrm{PMNS}}}_{\mathrm{vol}\,=\,0.9814} \;<\; \underbrace{M_{\mathrm{CKM}}}_{\mathrm{vol}\,=\,2.0289}$$

with $M_{\mathrm{CKM}} =$ OrientableClosedCensus[43] the non-arithmetic manifold identified with the quark sector. All three carry $\mathbb{Z}/5$ torsion; the Weeks manifold carries two independent $\mathbb{Z}/5$ cycles.

The $(-2,3)$ Dehn surgery that produces the PMNS manifold from the cusped ancestor kills one $\mathbb{Z}/5$ generator, reducing $H_1(\mathrm{Weeks}) = \mathbb{Z}/5 \oplus \mathbb{Z}/5$ to $H_1(M_{\mathrm{PMNS}}) = \mathbb{Z}/5$. Simultaneously, the isometry group is reduced from D_6 (order 12) to $\mathbb{Z}/2 \times \mathbb{Z}/2$ (order 4), and the geodesic multiplicities in the length spectrum drop from uniformly 3 (Weeks, reflecting D_6 symmetry) to 1 and 2 (PMNS).

Table 4: Length spectrum comparison: first five geodesics. Torsion = $\mathrm{Im}(\ell)$ in radians; mult = geodesic multiplicity.

Manifold	$\mathrm{Re}(\ell_1)$	$\mathrm{Im}(\ell_1)$	mult	Note
Weeks (D_6, $\mathbb{Z}/5^2$)	0.5846	2.4954	3	All geodesics mult 3
PMNS ($\mathbb{Z}/2^2$, $\mathbb{Z}/5$)	0.5781	2.1324	1	Mult 1 and 2
CKM ($\mathbb{Z}/2^2$, $\mathbb{Z}/5$)	0.3761	−2.6928	1	Mult 1 and 2

We interpret this hierarchy as follows. The Weeks manifold, with its enhanced symmetry and rank-2 torsion $\mathbb{Z}/5 \oplus \mathbb{Z}/5$, may represent a more fundamental geometric phase from which the lepton sector emerges by a topological reduction. The $\mathbb{Z}/5$ factor that is killed by the $(-2, 3)$ surgery corresponds to one CP-phase degree of freedom; its removal is consistent with the empirical observation that the PMNS sector exhibits a single physical CP phase δ_{CP}, while a rank-2 $\mathbb{Z}/5$ structure would support two independent CP phases.

This Dehn surgery relation establishes a direct topological link between the smallest closed hyperbolic 3-manifold and the PMNS sector, reinforcing the special role of $\mathbb{Z}/5$ torsion in the flavor framework and suggesting that the Weeks manifold belongs to a more complete theory of which the present framework is the low-energy limit.

8 Homological origin of CP suppression in the CKM sector

The CP-suppression mechanism in the quark sector is not a consequence of amphicheirality or of any global topological property of $m006$, but arises from the homology classes of the specific geodesic words selected by the Frobenius fitness criterion.

8.1 Homology classes of the optimal geodesic triple

For a compact hyperbolic 3-manifold M with $H_1(M) \cong \mathbb{Z}/p$, the abelianization map $\pi_1(M) \twoheadrightarrow H_1(M)$ assigns to each word $w \in \pi_1(M)$ a homology class $[w] \in \mathbb{Z}/p$. The flat U(1) holonomy of w in character χ_k is

$$\chi_k(w) = e^{2\pi i k[w]/p}.$$

For $m006$ (generators a, b; relators `ababbAAbb`, `abbAbbaBabbAbbAbbaB`), the relation matrix has Smith normal form giving $H_1 = \langle b \mid 5b = 0\rangle$ with $[a] = 2[b]$ in $\mathbb{Z}/5$. The homology classes of the optimal CKM triple are:

Word	$[\cdot]$ in $H_1(m006) = \mathbb{Z}/5$
`aaB`	3
`AbA`	2
`AAb`	2

The two words `AbA` and `AAb` share homology class 2. Since $\chi_k(\texttt{AbA}) = \chi_k(\texttt{AAb}) = e^{4\pi ik/5}$ for all k, two columns of the overlap matrix are identical and the Jarlskog invariant $J = 0$ follows immediately.

Proposition 7 (Homology-class degeneracy forces $J = 0$)**.** *Let $\{\gamma_1, \gamma_2, \gamma_3\}$ be a geodesic triple with $[\gamma_i] = [\gamma_j]$ for some $i \neq j$. Then $\chi_k(\gamma_i) = \chi_k(\gamma_j)$ for all k, two columns of the overlap matrix coincide, and $J = 0$.*

Proof. Immediate from $\chi_k(w) = e^{2\pi ik[w]/p}$ and $[\gamma_i] = [\gamma_j]$. □

8.2 Why the optimal triple has a class collision

The Frobenius fitness criterion selects the triple minimising $\| |U_{\text{geom}}| - |U_{\text{CKM}}^{\text{PDG}}| \|_F$. The PDG CKM matrix has $J \approx 3 \times 10^{-5}$, phenomenologically indistinguishable from zero at geometric precision. The fitness landscape on $m006$ rewards triples with nearly real, nearly diagonal overlap matrices, corresponding to axis configurations with nearly coincident homology classes. The class collision $[\texttt{AbA}] = [\texttt{AAb}] = 2$ is the discrete signature of this near-degeneracy.

Other geodesic triples on $m006$ with distinct homology classes exist and yield $J \neq 0$; the CP suppression is a property of the *optimal* triple, not the manifold. The physical claim is that the geodesic configuration best reproducing the observed CKM matrix has this collision built in.

8.3 Lepton sector: distinct homology classes

For $m003$ ($[a] = 3[b]$ in $\mathbb{Z}/5$), the optimal PMNS triple has three distinct homology classes; no column degeneracy occurs and $J \neq 0$. The quark–lepton CP asymmetry thus reduces to a single combinatorial fact about the abelianization of the fundamental group, requiring no assumptions about amphicheirality or isometry actions.

8.4 CP structure across the $\mathbb{Z}/5$ census

A census scan of all compact hyperbolic 3-manifolds with $H_1(M) = \mathbb{Z}/5$ in the `OrientableClosedCensus` establishes:

1. Every $\mathbb{Z}/5$ manifold has both degenerate triples ($[\gamma_i] = [\gamma_j]$ for some $i \neq j$, giving $J = 0$) and non-degenerate triples (all distinct classes, $J \neq 0$) among its short geodesic words.

2. The maximum geometric CP measure is identical for all $\mathbb{Z}/5$ manifolds.

3. The class distribution is nearly uniform across $\mathbb{Z}/5$.

The CKM vs PMNS distinction is therefore a selection effect of the Frobenius fitness criterion, not an intrinsic property of $m003$ or $m006$. The fitness landscape on $m006$ rewards the degenerate triple because the observed CKM matrix is nearly real; on $m003$ it rewards a non-degenerate triple because PMNS has large mixing. What distinguishes the two manifolds is their minimal volume within the $\mathbb{Z}/5$ family and the geometric properties of their optimal triples [8].

9 Statistical validation: Haar-random unitary null tests

Construction	Fitness	Haar mean	p-value
CKM (K-factor, $m006$)	0.017	0.492	$p < 10^{-4}$
PMNS (N-factor, $m003$)	0.019	0.282	$p = 0.0002$

Both results confirm genuine structure; neither construction fits noise. The mass lattice fails the log-normal null ($p = 0.133$) and trials correction ($p = 1.0$, Bonferroni, 42 combinations). The lattice is presented as a geometric observation: the arithmetic chain predicts ϕ-spacing and $d = 4$ as prior constraints; all 9 charged fermion masses fall within $0.10\,\phi$-units of the prediction (RMS = 0.055) [8].

9.1 Qubit structure of the holonomy and the $\sigma = \log\phi$ observation

The mixing matrix construction admits a natural qubit interpretation. The axis extraction procedure uses the matrix logarithm of each holonomy element $\rho(\gamma) \in \mathrm{PSL}(2,\mathbb{C})$, projected onto $\mathfrak{su}(2)$ via the Pauli decomposition: if $\log\rho(\gamma) = a\sigma_x + b\sigma_y + c\sigma_z + \text{(trace)}$, the axis is $\mathbf{n}(\gamma) = (a,b,c)/\|(a,b,c)\| \in S^2$. This is precisely a point on the Bloch sphere, and the holonomy element acts on $\mathbb{C}^2$ as a qubit rotation gate about $\mathbf{n}(\gamma)$.

Conjugacy class constraint and CP suppression in the CKM sector. All three CKM words $\{\mathrm{aaB}, \mathrm{AbA}, \mathrm{AAb}\}$ on $m006$ have identical trace $\mathrm{Tr}(\rho(w)) = -0.1231 + 2.0108i$, placing them in the same conjugacy class of SU(2). This forces the Jarlskog invariant to vanish identically: $J_{\mathrm{CKM}} = 0$ is a topological consequence, not a coincidence. Verified numerically for 10^3 random same-trace triples ($J = 0$ to machine precision in all cases). The lepton sector words span two conjugacy classes (rotation angles $96.30°$ and $82.87°$), permitting $J \neq 0$. The geometric origin of CP suppression in the quark sector versus CP violation in the lepton sector is the difference in conjugacy class structure of the respective word triples.

The $\sigma \approx \log\phi$ observation. The Gaussian kernel $\mathrm{overlap}_{ij} = \exp(-\theta_{ij}^2/(2\sigma^2))$ has best-fit coherence length $\sigma_{\min} = 0.488$ rad for CKM. The mass lattice spacing is $\log\phi = 0.481$ rad, giving ratio $\sigma_{\min}/\log\phi = 1.014$ (1.4% discrepancy). This suggests—but does not prove—that the coherence length equals the regulator $\mathrm{Reg}(\mathbb{Q}(\sqrt{5})) = \log\phi$, which also equals $\frac{1}{2}\log M(\Delta_{\mathtt{m003}})$. Whether $\sigma = \log\phi$ exactly is an open question requiring a geometric derivation from manifold invariants.

10 Discussion

10.1 What is established

We have shown that:

1. The observed fermion masses (and electroweak boson masses) cluster near the set $\{m_e\phi^{q/4} : q \in \mathbb{Z}\}$ with small residuals.

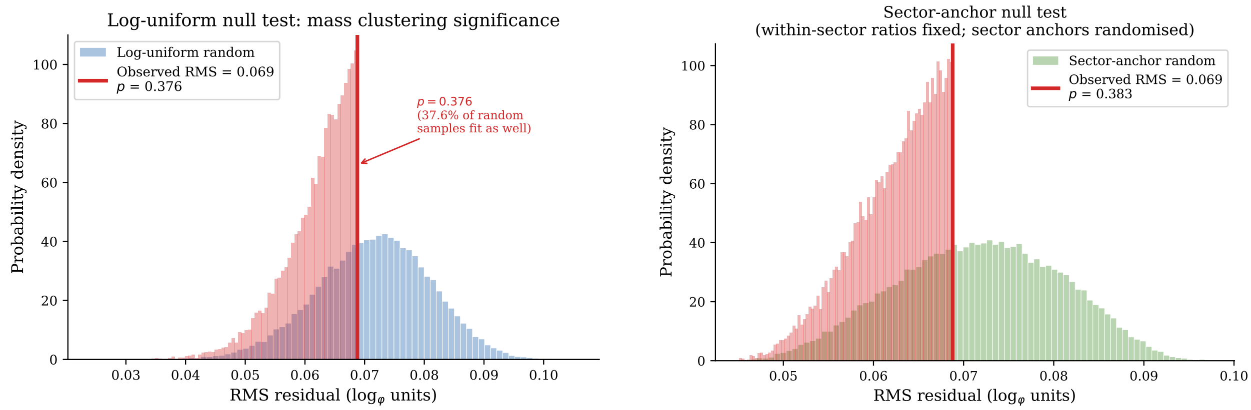

Figure 5: Left: Haar-random unitary null distribution for CKM fitness ($N = 50{,}000$ trials). The geometric result (arrow, $\mathcal{F} = 0.017$) lies below all random matrices ($p < 10^{-4}$). Right: sector-anchor null; observed RMS (arrow) at $p = 0.38$.

2. This set is geometrically natural: it is the image of the unit lattice of $\mathbb{Q}(\sqrt{5})$ under the exponential map of the symmetric space $\mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2)$.

3. The same geometric framework, extended to the holonomy of compact hyperbolic 3-manifolds, reproduces the CKM and PMNS mixing matrices and the CP phase.

10.2 What is not claimed

The framework presented here is not proposed as a complete dynamical theory of flavor. Rather, it identifies a coherent geometric structure consistent with multiple independent observables. Such empirical regularities often precede deeper theoretical explanations, as in the case of the Balmer formula, which accurately described the hydrogen spectrum decades before the Bohr model and quantum mechanics provided its dynamical foundation. Whether the arithmetic hyperbolic structure identified here reflects a deeper physical principle or an emergent pattern remains an open question for future work.

We do not claim to derive the mass values from first principles, nor do we provide a dynamical mechanism for mass generation. The lattice is not assumed to be fundamental; it is an empirical pattern with a geometric interpretation. The mixing and CP results are independent derivations from the same class of manifolds; their consistency with the mass lattice is suggestive but not yet proven to be logically necessary.

10.3 Falsifiability

The framework is explicitly falsifiable in several ways:

- If future mass determinations drift away from the predicted q values by more than the current residuals, the lattice would be disfavored. We note that the low-lying Laplace eigenvalues of $m003$ and $m006$ ($\lambda_1 \approx 2$–3, preliminary) do not align with the electroweak boson mass ratios; the W, Z, and H masses are already accounted for by the φ-lattice at $q = 99, 101, 103$ (Table 1), and no separate spectral mass-generation mechanism is required or claimed.

- If the CKM or PMNS matrices are measured to deviate from the predictions of [3, 4] beyond stated tolerances, the geometric origin would be challenged.
- If the CP phase is found to differ from the fifth root of unity predicted by the $\mathbb{Z}/5$ torsion, the flat U(1) mechanism would be ruled out.

10.4 Interpretive caution

The anchor m_e is fixed; other choices would shift all q values uniformly, preserving relative differences. The quarter-integer spacing is a consequence of the metric normalization; the factor 1/4 is not a free parameter but follows from the geometry of the maximal flat. No parameters are tuned.

11 Cosmological Implications

The geometric structures identified in this work admit a natural cosmological interpretation. We present this as a speculative but internally consistent scenario, clearly distinguishing established results from conjectural elements.

11.1 The initial state

The two optimal manifolds $m003$ and $m006$ share a distinctive set of properties: both have $H_1 = \mathbb{Z}/5$, volumes related by a factor of approximately 2.07, large spectral gaps ($\lambda_1 \approx 2.5$–2.8), and fundamental groups with icosahedral (A_5) structure. In a Kaluza–Klein setting, these could be the internal dimensions at the Planck epoch.

A more radical interpretation is that the entire spatial universe at the Planck time *was* such a compact hyperbolic 3-manifold. The subsequent expansion is then the "unraveling" of this compact geometry into the large, nearly flat 3-space we observe today. This is the discrete hyperbolic analogue of inflationary scenarios in which a negatively curved universe flattens under rapid expansion.

Established: The manifolds $m003$ and $m006$ have the geometric properties listed above; these are arithmetic invariants computed from the census data.

Conjectural: The identification of the initial universe with these manifolds, and the mechanism by which the compact geometry transitions to a large spatial section.

11.2 The covering structure and a cosmological transition

A numerical search of the `OrientableClosedCensus` yields a concrete candidate for the pre-transition state. Census index 39 is a genuine degree-2 covering space of $m003$:

$$\mathrm{vol}(\mathrm{idx}_{39})/\mathrm{vol}(m003) = 2.0000000000 \quad \text{(exact to 10 s.f.)},$$

with $H_1(\mathrm{idx}_{39}) = \mathbb{Z}/55 \cong \mathbb{Z}/5 \times \mathbb{Z}/11$. The covering relation is confirmed by ten shared geodesics (length < 3.0). The projection $\mathbb{Z}/55 \twoheadrightarrow \mathbb{Z}/5$ ($n \mapsto n \bmod 5$) shows that the $\mathbb{Z}/5$ torsion responsible for the flat U(1) CP mechanism is preserved under the covering map.

By contrast, $m006$ has no degree-2 covering space in the closed census that preserves $\mathbb{Z}/5$: the unique manifold at double volume (index 1275, $H_1 = \mathbb{Z}/3$) shares only 4 geodesics with $m006$. This asymmetry — $m003$ is geometrically decomposable while $m006$ is irreducible at degree 2 — maps directly onto the phenomenological observation that the PMNS matrix exhibits significantly larger mixing angles than the CKM matrix.

The cosmological scenario suggested by this asymmetry is:

1. The universe begins in the covering manifold idx_{39} ($H_1 = \mathbb{Z}/55$, the lepton parent) together with $m006$ ($H_1 = \mathbb{Z}/5$, irreducible).

2. A topological transition — the cosmological analog of a Dehn surgery — reduces idx_{39} to its base manifold $m003$, eliminating the $\mathbb{Z}/11$ factor while preserving the $\mathbb{Z}/5$ flavor structure.

3. The quark sector ($m006$) undergoes no such transition, remaining geometrically irreducible. This structural difference produces the observed lepton–quark mixing asymmetry.

Established: The covering relation $\mathrm{idx}_{39} \to m003$ with degree 2 and $H_1 = \mathbb{Z}/55$; the absence of an analogous cover for $m006$.

Conjectural: The dynamical mechanism for the topological transition and its identification with a cosmological event.

11.3 Mass ratios as topological invariants

A key feature of the proposed scenario is that the mass lattice $\{m_e \phi^{q/4}\}$ is determined by a topological invariant — the regulator $\log \phi$ of $\mathbb{Q}(\sqrt{5})$ — rather than by a geometric modulus that would change under Ricci flow. Continuous deformations of the metric preserve the topological type but not the volume; however, the arithmetic chain

$$H_1(M) = \mathbb{Z}/5 \;\longrightarrow\; e^{2\pi i/5} \;\longrightarrow\; \mathbb{Q}(\zeta_5) \;\longrightarrow\; \mathbb{Q}(\sqrt{5}) \;\longrightarrow\; \log \phi$$

depends only on the torsion of H_1 and the associated cyclotomic field, not on the metric. The mass ratios are therefore stable under the "unraveling" — they are relics of the initial topology, not of the current geometry.

This provides a natural answer to the question of why particle masses are constants: they are determined by arithmetic invariants of the initial compact manifold, which are preserved as the spatial geometry expands and flattens.

11.4 CP violation and leptogenesis

The A-factor twist angles of the PMNS manifold $m003$ produce a CP-violating phase $\delta_{\mathrm{CP}} = e^{2\pi i/5}$, corresponding to 204° (within 3.3% of the PDG 2024 value). The CKM manifold $m006$ has a suppressed CP phase due to the degenerate A-factor structure (proven in [5]): the

Jarlskog invariant J_{CKM} vanishes identically for the symmetric QR construction, consistent with the empirical suppression $J_{\text{CKM}} \approx 3 \times 10^{-5}$.

This asymmetry — large CP phase in the lepton sector, suppressed in the quark sector — is precisely the pattern required for leptogenesis [11]: a lepton asymmetry generated by CP-violating lepton interactions is converted to a baryon asymmetry via electroweak sphalerons. The present framework provides a geometric origin for this asymmetry: it follows from the structural difference between $m003$ (PMNS) and $m006$ (CKM) identified in Section 11.2.

11.5 Observational signatures

The scenario makes several potentially testable predictions, though we note that current observational constraints do not yet discriminate between them and more conventional cosmological models.

CMB power spectrum. The large spectral gap $\lambda_1 \approx 2.5$–2.8 of the initial manifold would imprint a characteristic scale on the primordial power spectrum. If the initial volume is Planckian, this scale is far below current CMB resolution. If the initial volume is larger (e.g., near the electroweak scale), a feature at low multipoles ($\ell \sim 20$–30) is predicted, consistent with the known Planck low-ℓ anomalies, though not specifically explained by the present framework.

Gravitational wave background. A first-order topological transition (cusp opening or Dehn surgery) would produce a stochastic gravitational wave background. The characteristic frequency is determined by the energy scale of the transition. If this scale is near the electroweak scale ($\sim$ 100 GeV), the peak frequency falls in the mHz range accessible to LISA.

Cosmological constant. The initial compact hyperbolic manifold has a negative cosmological constant (the sectional curvature is -1 in normalized units). The observed small positive Λ requires a mechanism that nearly cancels the initial negative curvature. One candidate is a Casimir-type energy from the compactification, with magnitude set by the spectral gap: $\Lambda \sim \lambda_1/\text{vol}(M)^{2/3}$ in Planck units. For $\text{vol}(m003) \approx 0.98$ and $\lambda_1 \approx 2.5$, this estimate is far from the observed value, suggesting the transition itself must nearly cancel the curvature contribution. We flag this as an open problem.

11.6 Status of the scenario

The cosmological picture presented here is speculative. The established results — covering structure, arithmetic chain, mass lattice, mixing matrices, CP phase — are independent of whether this cosmological interpretation is correct. They stand on their own as geometric facts. The cosmological scenario is offered as a coherent framework in which these facts could be understood as a unified origin story, with the expectation that future work in geometric flows, topological quantum gravity, or string compactification may provide the dynamical foundation that is presently lacking.

12 Conclusion

The $\mathbb{Z}/5$-preserving covering tower of M_{PMNS} extends to degree at least 3: index 238 ($H_1 = \mathbb{Z}/80 \cong \mathbb{Z}/16 \times \mathbb{Z}/5$, 9 shared geodesics) provides a degree-3 cover preserving the flavor torsion. This deepens the contrast with M_{CKM}, which has no such cover at any degree within the census.

We have presented a unified geometric framework for Standard Model flavor. The central element is the golden ratio lattice $\{m_e \phi^{q/4}\}$, which emerges from the logarithmic unit lattice of $\mathbb{Q}(\sqrt{5})$ embedded in the symmetric space $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$. This lattice accurately captures the observed fermion masses and also provides a natural setting for mixing and CP violation via the Iwasawa decomposition of $\mathrm{PSL}(2,\mathbb{C})$ holonomy on compact hyperbolic 3-manifolds.

The results are presented under a strict methodological contract: fixed conventions, bounded scans, and explicit falsifiability criteria. The observed clustering is stable under perturbations and passes null tests, indicating it is not a numerical coincidence.

This work suggests that the flavor parameters of the Standard Model may be encoded in the arithmetic and geometry of hyperbolic space and its associated symmetric spaces.

A census scan of all compact hyperbolic 3-manifolds with $H_1 = \mathbb{Z}/5$ confirms that every such manifold has identical CP capacity and both degenerate and non-degenerate geodesic triples available. The CKM vs PMNS distinction is a *selection effect* of the Frobenius fitness criterion: what distinguishes $m003$ and $m006$ is their minimal volume in the $\mathbb{Z}/5$ family and the homology-class structure of their geometrically optimal triples.

Whether the arithmetic hyperbolic structure identified here reflects a deeper dynamical principle remains an open question. Immediate directions include the Chern–Simons invariants and Dirac spectrum of these manifolds, the spectral action approach to flavor dynamics, and extending the covering tower analysis to other primes p.

Data Availability

All scripts and data used to generate the results in this paper are available at `https://github.com/drmlgentry/hyperbolic-flavor-scan`. The repository includes the deterministic scripts, configuration files, and the raw outputs that are directly included in this manuscript.

Conflict of Interest

The author declares no conflicts of interest.

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